

ROAD-05-CERTIFICATION-REGULATORY-PATHWAY

ROAD-05: CERTIFICATION & REGULATORY PATHWAY

Carrington Storm Motors — Project AEGIS

UL, FCC, NIST, MIL-STD, IEC, ISO, Building Code, Maritime Class — Complete Regulatory Roadmap

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EXECUTIVE SUMMARY

Every CSM product must navigate a multi-jurisdictional regulatory gauntlet before commercial deployment. This document maps the complete certification and regulatory pathway for all 50+ CSM products across 8 regulatory domains: electrical safety (UL), communications (FCC), electromagnetic resilience (NIST/MIL-STD), international standards (IEC/ISO), building codes (ICC/IAPMO), maritime classification (ABS/DNV/Lloyd’s), automotive (IATF/FMVSS), and materials (ASTM). Each regulatory pathway is detailed with required standards, testing laboratories, estimated timeline, estimated cost, and dependency mapping. Products are grouped by regulatory similarity to maximize shared certification testing and minimize total compliance cost.

The Regulatory Philosophy: CSM products are novel — no existing certification category exists for “Carrington-hardened residential building” or “GIC-immune humanoid robot.” CSM will work proactively with standards bodies to define the certification categories, positioning itself as the standard-setter rather than a late follower. Where no standard exists, CSM will publish its test protocols as proposed industry standards via IEEE, ASTM, and NIST working groups.

REGULATORY DOMAIN 1: ELECTRICAL SAFETY — UL CERTIFICATION

1.1 UL Standards Applicable to CSM Products

Product	UL Standard	Test Description	Lab	Timeline	Cost
Aegis Home Wall Panel	UL 94	Flammability of plastic materials (V-0 required for building materials)	UL Solutions, Northbrook IL	3 months	\$12K
Aegis Home Wall Panel	UL 723 (ASTM E84)	Surface burning characteristics (Flame Spread Index <25, Smoke Developed Index <450)	UL Solutions	2 months	\$8K
Aegis Home Wall	UL 746C	Outdoor weathering (UV, moisture, thermal	UL Solutions	4 months	\$22K

Panel		cycling) — 2,000 hours Xenon arc			
Aegis Home Wall Panel	UL 263 (ASTM E119)	Fire resistance of building construction (1-hour rating for residential walls)	UL Solutions	4 months	\$28K
Safe-Wave Microwave	UL 923	Microwave cooking appliances	UL Solutions	3 months	\$18K
Stellar-Chill Refrigerator	UL 250	Household refrigerators and freezers	UL Solutions	3 months	\$16K
Carrington-Proof Oven	UL 858	Household electric ranges	UL Solutions	3 months	\$18K
Inflatable Ceramic Water Heater	UL 174	Household electric storage tank water heaters	UL Solutions	3 months	\$15K
Inductive-Proof Wiring System	UL 83 (modified)	Thermoplastic-insulated wires (PMMA POF — requires custom evaluation as optical fiber)	UL Solutions	6 months	\$45K
Non-Conductive Motor	UL 1004 series	Electric motors — rotating electrical machines	UL Solutions	4 months	\$22K
Aegis Home Complete Electrical System	UL 508A	Industrial control panels (for home energy management panel)	UL Solutions	2 months	\$10K
Phantom MK-1 Battery System	UL 1973	Batteries for stationary applications (LiFePO ₄)	UL Solutions	4 months	\$28K
Phantom MK-1 Complete Robot	UL 1740	Robots and robotic equipment (safety of autonomous systems)	UL Solutions	6 months	\$55K
Ascension Drone Battery	UL 1642	Lithium batteries (solid-state Toyota battery)	UL Solutions	4 months	\$22K
Protonic Transceiver	UL 60950-1 (transitioning to UL 62368-1)	Safety of information technology equipment	UL Solutions	3 months	\$18K
Aegis Embark Products (8) — Electrical Components	UL 2089 (modified)	Vehicle battery adapters / in-vehicle electronics	UL Solutions	4 months	\$35K

UL Certification Subtotal (16 product categories): \$392K, 48 months total (parallelized across products)

1.2 UL Certification Strategy

1. **Pre-Submission Meeting:** Schedule preliminary review with UL Solutions to introduce CSM’s non-metallic,

dielectric product architecture. Many UL standards assume metallic enclosures and conductive grounding — CSM’s products challenge these assumptions. Pre-submission avoids 6-month rework cycles.

2. **Custom UL Outline of Investigation:** For products with no exact standard match (Inductive-Proof Wiring via PMMA POF, Non-Conductive Motor), request a UL Outline of Investigation — a custom test plan developed jointly by CSM and UL engineers. This becomes the de facto standard for future dielectric electrical products.
3. **Witnessed Testing:** All TEM cell and GIC testing will be witnessed by UL engineers at CSM’s El Segundo facility to accelerate certification.

REGULATORY DOMAIN 2: COMMUNICATIONS — FCC COMPLIANCE

2.1 FCC Regulations Applicable

Product	FCC Rule Part	Description	Test Lab	Timeline	Cost
LoRa Mesh Radio (EMF Detector, LandOLil sensor nodes)	Part 15.247	Intentional radiators in 902-928 MHz ISM band (frequency hopping spread spectrum)	EMC test lab (TÜV SÜD or compatible)	3 months	\$12K
Protonic Transceiver	Part 15.209 (or exemption)	Intentional radiator, but protonic signaling is NOT RF — CSM will petition FCC for a declaratory ruling that protonic communication is not an intentional radiator under Part 15 because it uses proton transport, not electromagnetic wave propagation	FCC Office of Engineering and Technology	12 months (petition + ruling)	\$85K (legal + engineering)
PMMA POF Data Network	Part 15, Class A/B	Unintentional radiator (digital device) — POF network may emit minimal RF from optical-to-electrical converters at endpoints	EMC test lab	2 months	\$8K
Phantom MK-1 Onboard Electronics	Part 15, Class A	Industrial/scientific unintentional radiator	EMC test lab	2 months	\$10K
Ascension Drone Telemetry	Part 15.247 (or Part 87 for aviation)	Intentional radiator — may fall under Part 87 (aviation services) if used in controlled airspace; coordination with FAA required	EMC test lab + FAA coordination	6 months	\$35K
FEATHER LoRa Mesh Module	Part 15.247	Pre-certified module approach — use pre-certified LoRa radio module (Semtech SX1276-based) to avoid full intentional radiator	Module vendor certification + CSM integration testing	2 months	\$5K

testing

FCC Compliance Subtotal: \$155K, 24 months (parallelized)

2.2 FCC Strategy — Protonic Exemption

The Protonic Communication Network is the world’s first non-RF communication system intended for widespread deployment. FCC rules are written for electromagnetic radiation. Protonic signaling via hydrogen ion transport through Nafion membranes and BaZrO₃ solid-state conductors does not generate electromagnetic waves — it is an entirely different physical mechanism.

CSM will petition the FCC for a Declaratory Ruling that: 1. Protonic communication devices are not “intentional radiators” under 47 CFR §15.3(o) because they do not generate and emit radio frequency energy by radiation, conduction, or induction. 2. Protonic communication is not subject to Part 15 equipment authorization requirements. 3. In the alternative, if the FCC determines that incidental electric fields from proton transport constitute “unintentional radiation,” CSM requests classification under Part 15, Class A (industrial/commercial) with conducted and radiated emissions limits measured at standard distances.

Petition Strategy: Engage former FCC Chief Technologist as consulting expert. Submit physics analysis demonstrating proton mass = 1,836 × electron mass, proving electromagnetic radiation from proton transport is negligible below thermal noise floor at any distance >1 cm. Propose a new device classification: “Non-Electromagnetic Communication Device” (NECD) for FCC rules.

Fallback: If petition is denied or delayed, Protonic Transceiver will be certified as Part 15 Class A unintentional radiator (estimated compliance achievable with minimal shielding).

REGULATORY DOMAIN 3: ELECTROMAGNETIC RESILIENCE — NIST & MILITARY STANDARDS

3.1 NIST Standards Applicable

Standard	Description	CSM Product Relevance	Compliance Path	Timeline	Cost
NIST SP 800-53	Security and privacy controls — system hardening requirements	Phantom MK-1 cybersecurity, LandOLil data security	Self-attestation with third-party audit	4 months	\$25K (audit)
NIST SP 1800-series	Cybersecurity practice guides for critical infrastructure	Aegis Home energy management, infrastructure products	Conformance demonstration via NIST NCCoE	6 months	\$45K
NIST Framework for Improving Critical Infrastructure Cybersecurity	Risk management framework	All CSM infrastructure products	Self-assessment with independent validation	3 months	\$15K
IEEE P2945 (Draft)	Standard for GIC Hardening of Autonomous Systems	Phantom MK-1/MK-2 — CSM to participate in working group to influence standard development	Participate in IEEE working group; CSM test data to be submitted as reference implementation	Ongoing (standard target 2028 publication)	\$35K/year
MIL-STD-188-125	High-altitude EMP (HEMP) protection for ground-based C4I facilities	Phantom MK-1, Protonic Network infrastructure, military variants of Aegis Home	Testing at accredited HEMP test facility (White Sands or similar)	6 months	\$180K

MIL-STD-461	Electromagnetic interference characteristics of equipment	All CSM electronic products	CE102, CS101, CS114, CS115, CS116, RE102, RS103 testing	8 months (across all products)	\$220K
MIL-STD-810	Environmental engineering considerations (temperature, humidity, shock, vibration, salt fog)	Phantom MK-1, Ascension Drone, Infrastructure products	Full test suite per relevant methods	8 months	\$160K
MIL-STD-882	System safety	All CSM products with safety-critical functions	Safety assessment report, hazard analysis	3 months	\$35K
MIL-STD-1472	Human engineering design criteria	Phantom MK-1 human-robot interaction, Aegis Embark driver interfaces	Ergonomic assessment	2 months	\$18K

NIST/MIL-STD Compliance Subtotal: \$733K, 24 months

3.2 MIL-STD Strategy

CSM products are designed for civilian markets but built to military specifications. The MIL-STD-188-125 HEMP testing is the most demanding and expensive (\$180K, 6 months). CSM will: 1. Conduct HEMP testing only for products specifically marketed to defense customers (Phantom MK-1 military variant, hardened C4I shelters) 2. For civilian products, reference MIL-STD-461 and MIL-STD-810 test data as “surpasses military requirements” in marketing materials while noting that formal MIL-STD certification has not been obtained (avoids the cost and timeline of full certification) 3. Pursue formal MIL-STD certification under a Department of Defense contract vehicle (SBIR Phase III, DIU Other Transaction Authority, or DPA Title III) where the government funds the certification cost

REGULATORY DOMAIN 4: INTERNATIONAL STANDARDS — IEC & ISO

4.1 Applicable IEC Standards

Standard	Description	CSM Product	Timeline	Cost
IEC 61000-4-2	ESD immunity	All electronic products	1 month	\$5K (shared across products)
IEC 61000-4-3	Radiated RF immunity	All electronic products	1 month	\$8K
IEC 61000-4-4	Electrical fast transient/burst immunity	Aegis Home wiring, Phantom, Protonic	1 month	\$6K
IEC 61000-4-5	Surge immunity	Aegis Home, Infrastructure	1 month	\$6K
IEC 61000-4-6	Conducted RF immunity	All electronic products	1 month	\$6K
IEC 61000-4-8	Power frequency magnetic field immunity	ALL CSM PRODUCTS — core certification	3 months	\$45K (custom test setup for GIC simulation)
IEC 61000-4-9	Pulse magnetic field immunity	All products marketed for Carrington hardening	2 months	\$18K
IEC 61000-4-10	Damped oscillatory magnetic field immunity	Infrastructure, vessel products	2 months	\$15K

IEC 61000-4-11	Voltage dips and interruptions	Phantom, Protonic, Aegis Home energy management	1 month	\$5K
IEC 62368-1	Safety of audio/video and IT equipment (replacing 60950-1 and 60065)	Protonic Transceiver, LandOLil tablet	3 months	\$12K
IEC 60243	Dielectric strength of insulating materials	ALL CSM dielectric products — core certification	2 months	\$15K
IEC 60068	Environmental testing (temperature, humidity, vibration)	All products	4 months	\$35K

IEC Compliance Subtotal: \$176K, 12 months

4.2 ISO Certifications (Quality & Manufacturing)

Standard	Description	Scope	Timeline	Cost
ISO 9001:2015	Quality management system	CSM enterprise — prerequisite for all other certifications	12 months	\$85K
ISO 14001:2015	Environmental management system	CSM manufacturing operations	12 months	\$65K
ISO 45001:2018	Occupational health and safety	CSM manufacturing operations	12 months	\$55K
ISO 27001:2022	Information security management	LandOLil cloud infrastructure	8 months	\$48K
ISO 17025:2017	Testing and calibration laboratory competence	CSM QC laboratory (required for certified test data to be accepted by regulators)	18 months	\$120K
IATF 16949:2016	Automotive quality management (superset of ISO 9001)	Aegis Embark products for automotive OEM supply	18 months	\$95K
AS9100D	Aerospace quality management (superset of ISO 9001 + IATF 16949)	Ascension Drone (if pursuing defense contracts)	18 months	\$85K
ISO 13485:2016	Medical device quality management	Cervical-Guard (if pursuing medical device claims for neuro-entrainment), Way-Finder (if medical navigation claims)	12 months	\$65K

ISO Certification Subtotal: \$618K, 18 months (parallelized)

REGULATORY DOMAIN 5: BUILDING CODE APPROVAL PATHWAY

5.1 ICC Evaluation Service (ICC-ES) — United States

Product	ICC-ES Acceptance Criteria	Description	Timeline	Cost
Aegis-C Composite	AC04	Sandwich panels — acceptance criteria	14 months	\$85K

Wall Panel		for structural insulated panels		
BFRP Rebar	AC454	Fiber-reinforced polymer (FRP) bars for concrete reinforcement	12 months	\$65K
Geopolymer Concrete Foundation	AC470	Alternative cementitious materials for concrete	12 months	\$70K
Dielectric Door Hardware	AC156 (modified)	Door assemblies — custom evaluation path	10 months	\$45K
Aegis Home Complete Building System	AC481	Alternative building systems (custom evaluation for non-metallic residential construction)	18 months	\$120K

ICC-ES Subtotal: \$385K, 18 months

5.2 IAPMO (International Association of Plumbing and Mechanical Officials)

Product	IAPMO Standard	Description	Timeline	Cost
Conductive-Safe Pipe System (GFRP)	IGC 339 (or custom)	Non-metallic plumbing system evaluation	10 months	\$42K
Safe Stove Ventilation	UMC (Uniform Mechanical Code)	Mechanical exhaust systems	6 months	\$18K
Ceramic Water Heater	IAPMO/ANSI Z21.10.3	Gas water heaters (modified for electric ceramic)	8 months	\$28K

IAPMO Subtotal: \$88K, 16 months

5.3 California-Specific Approvals

Product	California Requirement	Description	Timeline	Cost
All Aegis Home Products	Title 24, Part 6 (Energy Code)	Energy efficiency compliance — Aegis Home aerogel insulation likely exceeds requirements	4 months	\$18K
Geopolymer Concrete	CALGreen (Title 24, Part 11)	Green building standards — geopolymer qualifies for LEED/GreenPoint Rated points	3 months	\$12K
YInMn Blue Exterior Coatings	SCAQMD Rule 1113 (VOC limits)	Architectural coatings VOC compliance — YInMn Blue is water-based, zero VOC formulation anticipated	2 months	\$8K
BFRP Structural Components	OSHPD (Office of Statewide Health Planning and Development) — seismic	Seismic approval for hospitals/schools (if targeting those markets)	12 months	\$55K

California Approvals Subtotal: \$93K, 12 months

5.4 International Building Code Recognition

Region	Code Body	Products	Timeline	Cost
European Union	ETA (European Technical Assessment) via EOTA	Aegis Home wall panels, BFRP rebar, geopolymer concrete	18 months	€95K (\$105K)
United Kingdom	BBA (British Board of Agrément)	Aegis Home, infrastructure products	14 months	£45K (\$58K)
Japan	BCJ (Building Center of Japan) — Ministerial Approval	Aegis Home (for earthquake-prone market)	18 months	¥12M (\$85K)
Australia/New Zealand	Codemark / WaterMark via ABCB	Aegis Home, GFRP pipe system	12 months	AU\$120K (\$82K)
Middle East	Dubai Central Laboratory / ADQCC	Aegis Home (for MENA market per Phase III)	12 months	\$55K

International Building Code Subtotal: \$385K, 18 months

REGULATORY DOMAIN 6: MARITIME CLASSIFICATION

6.1 Classification Society Approvals — Charlemagne Fleet

Classification Society	Scope	Requirements	Timeline	Cost
ABS (American Bureau of Shipping)	Steel Vessel Rules, Part 4 (Vessel Systems and Machinery), Part 5 (Specialized Vessels)	Hull structural review for BFRP retrofit, MXene coating compatibility with steel hull, LiFePO ₄ battery system safety (ABS Guide for Battery Propulsion), fire safety with non-metallic materials, emergency systems immunity verification	24 months (new technology qualification)	\$280K
DNV (Det Norske Veritas)	DNV-ST-0378 (Electrical energy storage), DNV-ST-0433 (Electrical power generation), DNV-CG-0194 (Alternative materials)	Dual class with ABS for European-flag vessels, PMMA POF data network qualification, dielectric fire suppression system approval	20 months	\$220K
Lloyd’s Register	LR Rules for Ships, Part 6 (Control, Electrical, and Fire)	UK-flag vessels, Charlemagne retrofit for UK cruise operators (Carnival UK, P&O)	18 months	£165K (\$215K)
Bureau Veritas	NR 320 (Safety of battery systems), NR 467 (Rules for steel ships — alternative materials)	French/Mediterranean-flag vessels (MSC, Costa, Ponant)	18 months	€185K (\$205K)

RINA (Registro Italiano Navale)	RINA Rules for Pleasure Yachts (modified for cruise)	Italian-flag vessels (Costa, MSC Italian fleet)	18 months	€165K (\$182K)
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Maritime Classification Subtotal: \$1.102M, 24 months

6.2 Maritime Regulatory Compliance (SOLAS/IMO)

Regulation	Requirement	Product Impact	Timeline	Cost
SOLAS Chapter II-1 (Construction — Structure)	Alternative design and arrangements for fire safety with non-metallic structural materials	Charlemagne BFRP structural retrofit — requires MSC/Circ. 1002 alternative design approval from flag state	12 months	\$85K
SOLAS Chapter II-2 (Fire Protection)	Fire test procedures for non-metallic equivalents (MSC.307(88) — FTP Code)	MXene-coated hull panels, BFRP deck elements, PMMA POF cable trays	8 months	\$65K
SOLAS Chapter III (Life-Saving Appliances)	Emergency communication devices must function after main power failure	Protonic Network as backup communication — qualification as “alternative life-saving communication”	6 months	\$35K
SOLAS Chapter IV (Radiocommunications)	GMDSS requirements — protonic communication as supplemental beyond GMDSS minimum	FCC + IMO coordination for protonic classification	10 months	\$45K
SOLAS Chapter V (Safety of Navigation)	FOG (fiber optic gyro) qualification as primary navigation sensor	Phantom navigation, Charlemagne bridge — requires type approval	14 months	\$120K
IMO Polar Code	Vessels operating in polar waters (if Charlemagne-class serves Arctic/Antarctic routes)	Low-temperature performance of MXene coating, LiFePO ₄ batteries at −40°C	8 months	\$55K
MARPOL Annex VI (Air Pollution)	Energy efficiency — Charlemagne LiFePO ₄ architecture may qualify for EEDI/EEXI improvement credit	Vessel Energy Efficiency Design Index improvement via electrification	4 months	\$15K

IMO/SOLAS Subtotal: \$420K, 14 months

REGULATORY DOMAIN 7: AUTOMOTIVE CERTIFICATION

7.1 Aegis Embark Product Certification

Product	Standard	Description	Timeline	Cost
All 8 Aegis Embark Products	IATF 16949:2016	Automotive quality management system (prerequisite for Tier 1 supply)	18 months	\$95K (included in ISO section)

All Electronic Components (piezos, MRE controllers)	AEC-Q200	Qualification of passive components for automotive stress	12 months	\$85K
Neural-Grip MRE Steering Wrap	FMVSS 124	Steering control systems — non-interference with airbag deployment, crashworthiness	8 months	\$48K
Cervical-Guard Headrest	FMVSS 202a	Head restraints — must meet existing head restraint standards with additional function	6 months	\$35K
Stellar-Tint Window Film	FMVSS 205	Glazing materials — visible light transmission must meet >70% VLT for windshield, >35% for side windows in most states	4 months	\$22K
Thorax-Calm Seatbelt Cover	FMVSS 209	Seat belt assemblies — STF cover must not impair webbing retraction, load limiting, or buckle release	6 months	\$38K
Mag-Float Seat Rail	FMVSS 207	Seating systems — seat must remain locked under crash loads; diamagnetic levitation must have positive mechanical lock	8 months	\$52K
Aegis Embark Complete Vehicle Integration	FMVSS 305	Electrolyte spillage and electrical shock protection — applies to EV high-voltage systems; CSM dielectric products must maintain isolation	6 months	\$28K
Aegis Embark — Aftermarket Install	NHTSA “Make Inoperative” prohibition (49 USC 30122)	Aftermarket installation must not disable any federally required safety equipment — CSM will submit Letter of Interpretation to NHTSA confirming Aegis Embark products do not “make inoperative” any safety system	8 months	\$35K (legal)

Automotive Certification Subtotal: \$438K, 18 months

REGULATORY DOMAIN 8: MATERIALS STANDARDS — ASTM

Material / Product	ASTM Standard	Description	Timeline	Cost
MXene EMI Coating	ASTM D357	DC resistance or	1 month	\$2K

MXene EMI Coating	ASTM D257	conductance of insulating materials	1 month	\$5K
MXene EMI Coating	ASTM D4935	Shielding effectiveness of planar materials (TEM cell method)	1 month	\$5K
BFRP Composite	ASTM D3039	Tensile properties of polymer matrix composites	1 month	\$4K
BFRP Composite	ASTM D7264	Flexural properties of polymer matrix composites	1 month	\$4K
BFRP Composite	ASTM D2344	Short-beam shear strength (interlaminar shear)	1 month	\$3K
BFRP Rebar	ASTM D7205	Tensile properties of FRP reinforcing bars	1 month	\$4K
BFRP Rebar	ASTM D7914	Long-term creep of FRP bars	6 months	\$18K
BFRP Rebar	ASTM D7957	Specification for solid round GFRP bars for concrete reinforcement	3 months	\$12K
Geopolymer Concrete	ASTM C39	Compressive strength of cylindrical specimens	1 month	\$3K
Geopolymer Concrete	ASTM C496	Splitting tensile strength	1 month	\$3K
Geopolymer Concrete	ASTM C1202	Electrical indication of concrete's ability to resist chloride ion penetration	1 month	\$4K
Ceramic Bearings (Si ₃ N ₄)	ASTM F2094	Standard specification for silicon nitride bearing balls	2 months	\$8K
Aerogel Insulation	ASTM C518	Steady-state thermal transmission properties (heat flow meter)	1 month	\$4K
YInMn Blue Coating	ASTM G154	Operating fluorescent UV and condensation apparatus (accelerated weathering)	3 months	\$12K
YInMn Blue Coating	ASTM B117	Salt spray (fog) apparatus	2 months	\$8K
MXene Spray Coating	ASTM D3359	Adhesion by tape test (cross-hatch)	1 month	\$2K
Dielectric Strength — All Products	ASTM D149	Dielectric breakdown voltage and dielectric strength	1 month	\$3K
All Composite Products	ASTM E84	Surface burning characteristics (see UL 723)	2 months	\$8K
PMMA POF Cable	ASTM D6577	Standard guide for testing industrial protective coatings (for MXene overcoat on POF)	1 month	\$3K

Smart Rope Actuator	ASTM D4268	Standard test methods for fiber ropes — modified for smart rope	2 months	\$6K
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ASTM Testing Subtotal: \$117K, 12 months (parallelized across materials)

CERTIFICATION COST AND TIMELINE — MASTER SUMMARY

Regulatory Domain	Total Cost	Total Timeline (Parallelized)	Critical Path?
1. UL Electrical Safety	\$392K	48 months	YES — Aegis Home cannot sell without UL
2. FCC Communications	\$155K	24 months	YES — for products with intentional radiators
3. NIST/MIL-STD Resilience	\$733K	24 months	NO — military certification is optional (revenue-enabling)
4. IEC/ISO Standards	\$794K	18 months	YES — ISO 9001 prerequisite for IATF, AS9100
5. Building Code (ICC-ES, IAPMO, California, International)	\$951K	18 months	YES — Aegis Home cannot sell without ICC-ES
6. Maritime Classification + SOLAS	\$1,522K	24 months	YES — Charlemagne Fleet cannot sail without class
7. Automotive (IATF, AEC-Q200, FMVSS)	\$438K	18 months	YES — Aegis Embark cannot supply OEMs
8. Materials (ASTM)	\$117K	12 months	NO — supportive role (data feeds other certifications)
TOTAL CERTIFICATION BUDGET	\$5,102K	48 months (2026-2030)	

REGULATORY DEPENDENCY MAP

Q3 2026: ASTM baseline testing → feeds all other certifications
Q4 2026: UL pre-submission meeting + ISO 9001 implementation start
Q1 2027: FCC Part 15 compliance (LoRa modules)
Q2 2027: UL 94, E84, 746C testing (Aegis Home panels) — 6-month testing window
Q3 2027: IEC 61000-4-8 magnetic field immunity (core CSM certification)
Q4 2027: ICC-ES AC04, AC454, AC470 submission (12–18 month evaluation)
Q1 2028: Protonic FCC petition filed
Q2 2028: ISO 9001 certification achieved (prerequisite for IATF 16949, AS9100D)
Q3 2028: Charlemagne ABS/DNV pre-engagement
Q4 2028: Aegis Embark AEC-Q200 + FMVSS testing begins (18 months)
Q1 2029: ICC-ES evaluation reports issued (Aegis Home can legally sell)
Q2 2029: IATF 16949 certification achieved (Aegis Embark can supply automotive OEMs)
Q3 2029: Charlemagne ABS new technology qualification complete; MIL-STD-188-125 HEMP testing
Q4 2029: ISO 17025 lab accreditation achieved
Q1 2030: Full certification portfolio complete — all products certified for all target markets

RISK REGISTER — CERTIFICATION DELAYS

Risk	Probability	Impact	Mitigation
ICC-ES evaluation exceeds projected 18 months	HIGH (30%)	6+ month delay to Aegis Home commercial sales	Parallel testing at multiple labs; hire former ICC-ES evaluator as consultant; begin evaluation process

FCC denies protonic exemption	MEDIUM (20%)	Protonic classified as intentional radiator, adding \$85K+ and 12-month testing	Q4 2026 (earliest possible) Fallback to Part 15 Class A certification (manageable)
ABS declines to classify Charlemagne without 5-year service history	MEDIUM (25%)	Fleet program delayed by years	Pre-engagement with ABS; submit CSM test data proactively; consider flag state "single voyage permit" for demonstration while pursuing full classification
IATF 16949 certification requires existing automotive supply history	MEDIUM (15%)	Aegis Embark cannot achieve Tier 1 status without partner	Partner with established Tier 1 supplier for initial program; CSM as Tier 2 initially, transitioning to Tier 1
ISO 17025 lab accreditation timeline underestimated	HIGH (30%)	QC lab test data not accepted by regulators until accredited	Contract accredited third-party labs for initial certification testing; CSM lab pursues accreditation in parallel
International building code recognition stalls	MEDIUM (25%)	Global expansion delayed	Prioritize US market (largest, fastest); use ICC-ES reports as leverage for international reciprocity agreements

"Certification is not bureaucracy. It is the physics of safety translated into legal language. Every test standard references a failure mode that killed someone. UL 94 exists because buildings burned. SOLAS exists because ships sank. FMVSS exists because vehicles crashed. The fastest path to market is not bypassing certification — it is designing to exceed every standard from the first prototype. The Dielectric Citadel does not just resist the Sun's electromagnetic assault — it earns the right to be sold by proving, in accredited laboratories, that it protects people from everything else too." — Chester, Director of Quality Assurance

Document generated 2026-07-15. All cost estimates are $\pm 30\%$ at P50 confidence and include testing laboratory fees, certification body review fees, and internal engineering labor for test preparation. Legal fees for regulatory petitions (FCC, NHTSA interpretations) included at estimated outside counsel rates. Timelines assume parallel processing where testing windows allow; serial dependencies (e.g., ISO 9001 before IATF 16949) may extend critical path if initial certification is delayed.